

Transcript (Evaluation 2 – Participant 1)

Interviewer (I): First, can you locate and pick up the blue device?

Participant (P): Uh... can I teleport? Oh, I can. Nice.

I: Yes, any of the blue devices will work.

P: I see three of them. Okay, I'll just grab this one. Got it.

I: Great. Now, please leave the blue box aside and find the Environmental Control panel.

P: Environmental control... is it on the floor or on the wall?

I: It's on the wall.

P: Oh my gosh, it's behind me! Okay, I've found it.

I: Can you change the light brightness?

P: Yes, switching to bright... and now low. Nice, I like this.

I: Good. Next, can you turn on the background audio?

P: Yes, I hear it. Oh, I like the music!

I: Finally, can you take the blue device into another room?

P: Hmm... this is a very busy room. Where's the entrance? Oh, it looks like a big cube here—maybe the door is on the other side.

P: Okay, I think I found it. Yes, here we go.

P: By the way, the music mute button is quite far away. Maybe you could add a quick shortcut—like pressing the joystick to mute.

P: Also, a short tutorial panel at the start would help, explaining the controls.

P: Another idea: instead of just a brightness slider, you could add presets like “daytime” and “nighttime,” or environment-matching lights, like “nature” or “office.”

P: For the audio, you could make it more 3D and tangible—like a DJ deck, where removing a disk mutes the music.

P: For lighting, maybe a little sun object that you pinch bigger or smaller to change intensity. That would feel more natural.

I: Thanks. We also plan to create three different rooms: a park, a studio, a café, and maybe a public space, so users can test the UI in different environments with different background music.

P: That sounds good. Some people will prefer quick, efficient controls, while others enjoy more leisurely, immersive ones. It would be nice to offer both options.

I: Thank you!

P: I really liked the ability to change light sensitivity. It makes the space feel more realistic.

Transcript (Evaluation 2 – Participant 2)

Interviewer (I): Can you see the scene now?

Participant (P): Yes, I can see it, but it doesn't look like your scene at first. Let me check... Okay, I think it's here. Maybe I need to sit in your position, because my direction is opposite to his.

I: That's fine, the avatar will move. Let's try again.

I: Okay, let's start. The first task is: please pick up a blue device and hold it in your hand.

P: With a hand gesture? Like this?

I: Yes, pick it up.

P: Do I need to press a button? Which one?

I: Try the front trigger.

P: It's a bit far away. Can I walk closer? Ah, this button works. I see, now I've lifted it.

I: Good. Now, look at the wall and find the light control. Please adjust the lighting to your preference.

P: Okay, I can see it... adjusting... yes, it works.

I: Great. Next, can you turn on the background music?

P: Let me try again... I didn't hear anything at first. Wait—now I can hear it. Okay, the music is on.

I: Can you also turn it off?

P: Yes, done.

I: Finally, try carrying the device into another room.

P: Into another room? Okay... let me look. The entrance isn't very obvious. But yes, I found an empty room. I can go in. Done.

I: Perfect. Do you have any feedback or suggestions?

P: Honestly, I don't have much understanding yet. During the whole process, I just followed the steps. I didn't notice big changes, but everything worked fine—no bugs, functions all ran normally. Maybe I would need more hints or explanation about the purpose.

I: The idea is that this prototype will eventually be placed into a "blue interface" scene, so we can test how lighting affects the experience in different environments.

P: Okay, I see. Right now I can't give specific ideas since it's just a sample.

I: In my previous IP, the environment switching was done with teleportation between locations like "studio" and "café." Which do you prefer—walking through space or teleporting?

P: Just now, I experienced walking into the next room. Previously you had teleportation. Honestly, both can work. With walking, I need to look for the entrance myself, but the entrance wasn't clearly marked, so I relied on guessing. The short distance helped—it wasn't too difficult. But if it's farther, maybe add clear landmarks or signs.

P: Also, I didn't realise the other room was supposed to be a café, because it wasn't visually obvious. If the design can add cues or indicators for room identity, that would help a lot.

Transcript (Evaluation 2 – Participant 3)

Interviewer 1 (I1): Okay, cool.

Interviewer 2 (I2): Please look around the scene. You should be able to see three different devices, like an iPhone, a laptop, etc. Try them.

Participant (P): I'm trying to move forward, but I can't. I tried teleporting and then used the controller—now I can control a bit. What do you want me to do?

I2: Please pick up one device. Choose a blue one and grab it.

P: Okay... and then?

I2: Next, please look at the light control on the wall.

P: On the line... to turn?

I2: Yes, use the slider to adjust the brightness.

P: I pointed at it and saw a small white dot. I dragged it and the light changed.

I2: Great. Now could you turn on the background music?

P: Do I also use that? I tried the grip... yes, now I hear some background music.

I2: Is it comfortable? You can also turn it off. Now, could you take the blue device into another room?

P: I want to move, but I can't. I'm stuck. I'm not moving.

I1: The camera may detect the position from far away...

I2: Maybe you need to adjust your orientation.

P: Yes, I had to face a certain direction. Now I can move a bit and grab it. You want me to go to another room?

I1: Yes, that's the main issue you're having. Try to look around.

P: Where's the other room? Is it behind the grey wall? I saw something... okay, I found the entrance. I can walk in now.

I2: Good. That's the end of the main tasks.

P: What is the blue device for?

I2: It represents a device for UI testing. We will put some images or interfaces on it.

P: I notice you have a desk or something?

I2: Yes, there are different environments: the first one is a studio, another is a café, and another is a public space.

P: Okay, but what is the application based on? What's the concept? Right now I can't tell—the objects look like placeholders.

I1: We didn't explain earlier.

I2: The idea is to test how different environments influence perception.

P: Is it for productivity, or another kind of app? Right now I can't relate it to Figma or any specific tool.

I2: It's connected to mental exercises in the semester project.

P: I'm still confused. Figma is usually for digital design, so I don't see why I need to carry a 3D object here.

I2: The devices represent real forms of laptops or phones, so digital designs can be placed on them in VR. That way, you can test readability under different environments, lighting, and audio conditions.

P: I see. My feedback:

- **Navigation:** I prefer teleportation over joystick movement. The joystick was too sensitive and caused slight motion sickness. Also, when I faced the wall, I couldn't move, which was confusing.
- **Interaction design:** I wouldn't recommend flat panels (like sliders and buttons on a wall) in XR. Those are things a flat screen already does. In VR, you should use spatial interactions.
- **Clarity of purpose:** Without explanation, I couldn't tell what the application was about. Even after hearing "Figma," it didn't feel related. The environment should better reflect the concept.

I2: In my IP1 I used a button to change environments. Do you think walking or teleporting is better?

P: I don't think adding a button or slider is valuable for XR. Instead, think about what can **only** be achieved in XR but not on flat screens—like using body movements or spatial interactions.

P: My recommendation:

- Incorporate more spatial interactions.
- Use body movements to enhance the experience.
- Avoid replicating flat UI controls.

I3: Okay, so for your testing plan—how will you measure task completion? Are you timing me?

I2: Yes, we will time each participant.

I1: Do you think finding the room was a successful task?

P: Honestly, I felt confused. Carrying the device into another room was just navigation, and there wasn't much purpose beyond finding an entrance.

P: Also, do you have a usability questionnaire or difficulty rating?

I2: Yes, we plan to include that.

P: One more technical question: how did you make the environment change? Did you write any scripts?

I2: Yes, for the lights we used a slider linked to light intensity values (0.2–2).

P: Was it linked to a game object?

I2: Yes, the light game object is referenced in the script. We used Unity slider components to adjust the light intensity in real time.

P: Okay, that makes sense. Thanks for explaining.

Transcript (Evaluation 2 – Participant 4)

Interviewer (I): Okay, let's start. First, please pick up the blue device and hold it.

Participant (P): Okay... got it.

I: Great. Put it aside for now. Next, go to the wall and adjust the light to the brightness you like.

P: Adjusting the brightness... yes, okay.

I: Good. Now, please try turning on the background music.

P: Hmm... I don't hear it at first... oh, now it's on.

I: If it's too loud, you can also turn it off. Next, please take the blue device and move into another room.

P: Alright... but the room feels a bit far. Okay, I'm in a new space. Is this the other room?

I: Yes, that's fine. That completes the tasks.

P (feedback): I noticed some issues:

1. When adjusting the light, the effect wasn't very clear. My phone (the blue device) didn't seem to be affected by the light. The lamp only lit one side of the room, so changing brightness didn't feel obvious.
2. The three objects (phone, iPad, laptop?) were unclear. I could guess what they were, but they didn't look obvious.

3. Walking to another room took too long and was slow. Also, I wasn't sure what the new room represented. The purpose of this task wasn't very clear.

I: Where do you think improvements could be made?

P:

- The light adjustment should be **global**, not just limited to one room. Otherwise, if I move to another room, the light doesn't apply.
- The background music should probably be **on by default**; users can choose to turn it off. With the music, the whole experience feels a bit more interesting.
- You should clarify the **overall purpose**. For example, if the screen or device should be visibly affected by the lighting, then make that clear and consistent.

Transcript (Evaluation 2 – Participant 5)

Interviewer (I): Okay, first please pick up the blue device.

Participant (P): I see it... I'll walk closer. Okay, I've picked it up.

I: Good. Put it down, then adjust the lighting to the brightness you like.

P: Adjusted. Looks fine.

I: Great. Now try turning on the background music.

P: Okay, it's on. I can hear it. If I want, I can also turn it off.

I: Perfect. Next, please take the blue device into another room.

P: Which device? The blue one? Okay. (Looks around) The room feels far... I found another room behind me. Yes, I can walk in now.

I: Okay, that completes the tasks. Do you have any feedback?

P: Honestly, the functions work fine, but there are some issues:

- The room and the furniture are all white, so it was hard to see the table when I entered the new room.
- Maybe it's because I set the brightness too high—the lighting effect didn't look clear. I had to reduce it to see better.
- The lamp seems only implemented in one room. Other rooms don't have light effects yet, so the experience felt incomplete.

I: Right, the lighting was only added in one room for now.

P: Another issue is picking up objects. I had to move very close to grab them. Ideally, you could implement a **raycast selection point**—so that I can highlight and pick up items at a distance, without confusion.

I: Good point.

P: Also, when moving between rooms, I couldn't see the entrances clearly from my angle. Maybe add visible doorways or markers at the room entrances.

I: In my IP1, I used teleportation between rooms. Do you think walking or switching by teleport is better?

P: I personally prefer teleport (switch), because it's quicker. But if you want realism, walking makes sense. If you use walking, you'll also need to design the roads and surroundings to look good.

I: Do you think the interaction design can be improved further?

P: Yes, for example, when using ray selection, add **haptic feedback** when an object is highlighted. That would make interactions clearer and more immersive.

I: Thank you.